Elodie Maignant

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Current position

Mar 2026 – present **Postdoctoral researcher**, *Danmarks Tekniske Universitet, Denmark.*
"Decoding the coordination of biological networks". PIs: Pia Nyeng, Aasa Feragen, Anna Calissano.

Education

Oct 2020 – Dec 2023 **PhD in Applied Mathematics**, *Université Côte d'Azur, France.*
Under the supervision of Xavier Pennec and Alain Trouvé.

Sep 2016 – Sep 2020 **Bachelor's and Master's degrees – Mathematics**, *ENS Paris-Saclay, France.*
2020 Master's degree in Mathematics, Vision, Learning (MVA).
2019 Master's degree in Higher Education in Mathematics with specialisation in Effective Algebra. Successful candidate to the Agrégation de Mathématiques (rank 66/308).
2017 Bachelor's degree in Mathematics.

Sep 2014 – Jul 2016 **Classe préparatoire en Mathématiques et Physique**, *Lycée Saint Louis, Paris.*
Intensive two-year study course in Mathematics and Physics preparing for the national entrance exams to mathematics and engineering university programs.

Research experience

Jan 2024 – Dec 2025 **Postdoctoral researcher**, *Zuse Institute Berlin, Germany.*
"Geometric learning for Single-Cell RNA velocity modelling". PI: Christoph von Tycowicz.

Oct 2020 – Dec 2023 **PhD student**, *Université Côte d'Azur, France.*
"Barycentric embeddings for geometric manifold learning". Under the supervision of Xavier Pennec and Alain Trouvé.

Apr 2020 – Sep 2020 **Master's thesis**, *ENS Paris-Saclay, France.*
"Data embedding and symmetric spaces with applications to molecular dynamics". Under the supervision of Alain Trouvé.

Apr 2018 - Jul 2018 **Visiting Student**, *Albert-Ludwigs-Universität Freiburg, Germany.*
"Statistical analysis of geometric shapes with applications to anthropology". Visiting Philipp Harms.

Jan 2017 - Jun 2017 **Bachelor's thesis**, *ENS Paris-Saclay, France.*
"Learning stochastic systems in high dimension". Under the supervision of Alain Trouvé.

Awards

2024 **2nd PhD prize**
PhD prize of the Université Côte d'Azur in Automatic, Signal Processing and Image Analysis.

2021 **2nd prize**
ICLR 2021 challenge for computational geometry & topology.

Travel grants

2025 **COST Action CaLISTA**, EUR 2400
Supporting a short-term scientific mission at University College London to work with Anna Calissano on dimensionality reduction for network-valued data.

Publications

Submitted papers

2026 **"VeloTree: Inferring single-cell trajectories from RNA velocity fields with varifold distances"**
Elodie Maignant, Tim Conrad, Christoph von Tycowicz

2025 **"Barycentric subspace analysis of network-valued data"**
Elodie Maignant, Alain Trouvé, Xavier Pennec, Anna Calissano

Journal articles

2021 **"Identification of the primary factors determining the specificity of the human VKORC1 recognition by thioredoxin-fold proteins"**, *International Journal of Molecular Sciences 22.2: 802.*
Maxim Stolyarchuk, Julie Ledoux, Elodie Maignant, Alain Trouvé, Luba Tchertanov

Conference proceedings

2025 **"Tree inference with varifold distances"**, *GSI'25, Springer.*
Elodie Maignant, Tim Conrad, Christoph von Tycowicz

2023 **"Riemannian locally linear embedding with application to Kendall shape spaces"**, *GSI'23. Springer.*
Elodie Maignant, Alain Trouvé, Xavier Pennec

2023 **"Towards quotient barycentric subspaces"**, *GSI'23, Springer.*
Anna Calissano, Elodie Maignant, Xavier Pennec

2021 **"Parallel transport on Kendall shape spaces"**, *GSI'21, Springer.*
Nicolas Guigui, Elodie Maignant, Alain Trouvé, Xavier Pennec

Preprints

2019 **"Approximation of Riemannian distances and applications to distance-based learning on manifolds"**
Philipp Harms, Elodie Maignant, Stefan Schlager

Others

2021 **"ICLR 2021 challenge for computational geometry & topology: Design and results"**, *ICLR 2021.*
Nina Miolane, et al.

Talks

June 2026 **Math in Aquitaine: Geometry, Shapes and PDEs – Invited talk**, *Bergerac, France.*
"Varifold metrics between spatial networks"

June 2026 **Curves and Surfaces 2026 – Invited talk**, *Saint-Malo, France.*
"Inferring single-cell trajectories from RNA velocity fields with varifold distances"

Dec 2025 **CFE-CMStatistics 2025 – Invited talk**, *London, UK.*
"Trajectory inference with varifold distances"

Oct 2025 **GSI'25 – Contributed talk**, *Saint-Malo, France.*
"Tree inference with varifold distances"

Jul 2025 **Math in Umbria: Geometry, Shapes and PDEs – Invited talk**, *Città di Castello, Italy.*
"RNA velocity fields and tree inference"

Feb 2025 **Infinite-dimensional Geometry: Theory and Applications – Invited talk**, *ESI, Austria.*
"Geometry of single-cell trajectories"

Aug 2024 **COMPSTAT 2024 – Invited talk**, *Giessen, Germany.*
"Barycentric subspace analysis of networks"

Aug 2024 **Math in Maine: Geometry, Shapes and PDEs – Invited talk**, *Andover, USA.*
"RNA velocity fields"

Jun 2024 **POPNets Workshop – Invited talk**, *Copenhagen, Denmark.*
"Barycentric subspace analysis of networks"

May 2024 **Geometric Sciences in Action – Poster**, *CIRM, France.*
"Barycentric subspace analysis of networks"

Jan 2024 **Workshop on Small Data Analysis – Invited talk**, *Berlin, Germany.*
"Barycentric subspace analysis of a set of graphs"

Dec 2023 **Seminar on Shape Analysis – Invited talk**, *Paris, France.*
"Intrinsic methods for manifold-valued data"

Nov 2023 **Workshop on Dimension Reduction – Contributed talk**, *Lyon, France.*
"Barycentric subspace analysis of sets of graphs"

Aug 2023 **GSI'23 – Contributed talk**, *Saint-Malo, France.*
"Riemannian locally linear embedding with application to Kendall shape spaces"

Aug 2023 **GSI'23 – Contributed talk**, *Saint-Malo, France.*
"Towards quotient barycentric subspaces"

Aug 2023 **Statistical Learning Theory Lab – Invited talk**, *Seoul National University.*
"Barycentric geometry on manifolds and application to non-Euclidean dimensionality reduction"

Jul 2023 **Math in the Mine: Geometry, Shapes and PDEs – Invited talk**, *Tende, France.*
"Geodesics of orbit spaces, affine mappings of simple manifolds and some related questions in barycentric geometry"

Sep 2022 **Introductory School on Geometry and Statistics – Poster**, *Cargèse, France.*
"Looking for invariance in Locally Linear Embedding"

Jun 2022 **Curves and Surfaces 2022 – Poster**, *Arcachon, France.*
"Looking for invariance in Locally Linear Embedding"

Oct 2021 **CJC-MA 2021 – Contributed talk**, *Palaiseau, France.*
"A generalisation of Locally Linear Embedding to manifold-valued data"

Aug 2021 **GTDAML 2021 – Contributed talk**, *Online*
"Visualisation of Kendall shape spaces with Geomstats"

Jul 2021 **GSI'21 – Contributed talk**, *Paris, France.*
"Parallel transport on Kendall shape spaces"

Teaching

Oct 2020 – Jun 2022 **Teaching assistant, in charge of tutorials**, *Université Paris-Saclay, France.*
Global Analysis, Topology and Differential Calculus.

Sep 2018 – Sep 2020 **Interrogatrice en classe préparatoire**, *Lycée Saint-Louis, Paris.*
Examiner in Mathematics for weekly oral interrogations in small groups.

EDI and Outreach

Oct 2026 – present **Women in Machine Learning (WiML) Mentoring Programme**, *Online.*
Acting as a mentor for a year to a post-graduate degree applicant. WiML is committed to advancing the careers of women and non-binary people studying and working in machine learning.

Juin 2024 & 2025 **Lange Nacht der Wissenschaften**, *Zuse Institute Berlin.*
Facilitated a workshop on data visualization and exploration. Every year, German universities and research institutes open their doors to the public for an evening

March 2023 **Rendez-vous des Jeunes Mathématiciennes et Informaticiennes**, *ENS Paris-Saclay.*
Gave a short lecture on shape analysis.

Oct. 2022 **Fête de la science**, *ENS Paris-Saclay.*
Facilitated a workshop on image falsification

Nov. 2018 & 2019 **Rendez-vous des Jeunes Mathématiciennes**, *ENS Ulm.*
Co-organized with Arthur Touati and Marc Abboud. Over a weekend, about twenty female high school students gather to do mathematics, discuss careers in mathematics with female researchers, PhD students, and engineers, and address gender stereotypes in the field

Languages

French Native

English Complete working knowledge *Cambridge English Advanced C1*

German General working knowledge

Miscellaneous

Personal Interest I am passionate about music and cinema. I have been singing and playing the viola and the saxophone since I was a very young age. I also practiced judo at a high level for years.